

Gearbox Design Workbench

Sanitized Academic Portfolio Report

Design of a Two-Stage Parallel Helical Gearbox for an Industrial Blower System

Public release prepared under the Zazu Nanami pseudonym. Identifying submission data, document metadata, and school-specific cover details were removed for public distribution.

Technical content from the report body is preserved in the following pages.

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1. Executive Summary

This project presents the detailed design and analysis of a two-stage parallel helical gearbox intended for an industrial blower application. The system is designed to convert a high-torque, low-speed input ($300 \text{ N} \times \text{m}$ 40 rpm) from an electric motor into a high-speed output (1200 rpm) required for aerodynamic fan operation, achieving a total speed ratio of 30:1.

The design process encompasses the comprehensive structural analysis of three steel shafts (Input Shaft A, Intermediate Shaft B, and Output Shaft C), considering static loads, fatigue failure theories, and deflection constraints. A distinguishing feature of this study is the development and utilization of a **custom Python-based Computational Design Tool (Gearbox Workbench)**. This tool facilitated rapid design iterations by automating the calculation of helical gear forces, generating precise Shear Force and Bending Moment Diagrams (SFD/BMD), and performing real-time fatigue safety checks based on the DE-Gerber criterion.

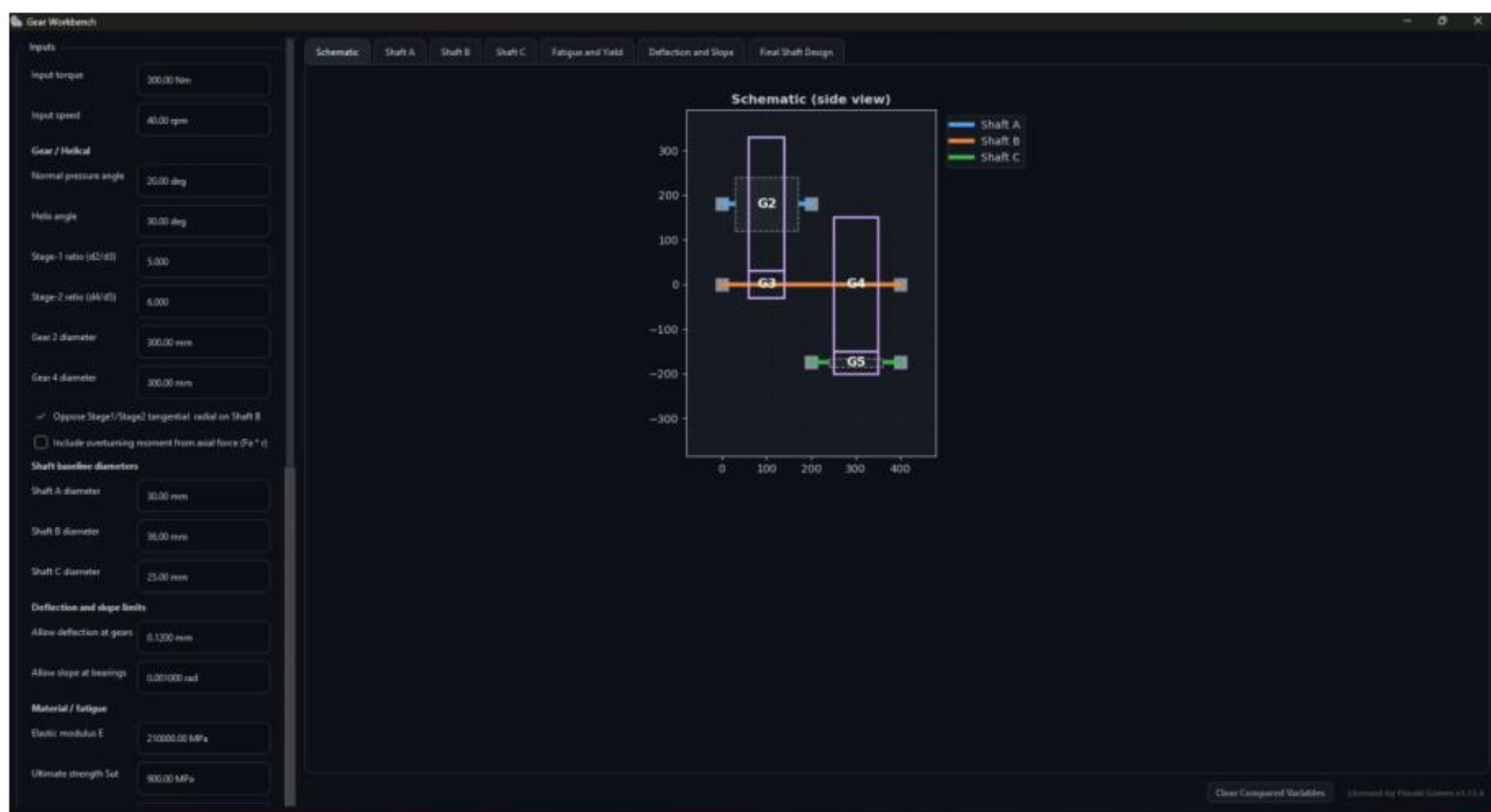


Figure 1: Graphical User Interface (GUI) of the developed Gearbox Workbench tool, showing the system schematic and real-time input parameters.

The final design successfully meets all safety requirements, with fatigue safety factors (n_f) exceeding the target of 2.0 and yielding safety factors (n_y) exceeding 3.0 for all shafts. The intermediate shaft (Shaft B) was identified as the critical component due to the complex loading from two meshing gears; however, the optimized design ensures that maximum deflections remain within the allowable limit of 0.12 mm (or user-defined limit) to guarantee proper gear meshing.

2. Introduction

2.1 Project Background and Objective

High-speed industrial ventilation systems require robust transmission mechanisms to step up the rotational speed of standard electric motors. The objective of this project is to design a reliable two-stage speed-increasing gearbox. The system must transmit $300 \text{ N} \times \text{m}$ of torque while ensuring structural integrity under continuous operation.

2.2 Design Specifications

The gearbox configuration consists of three parallel shafts supported by rolling-element bearings. The power transmission is achieved through two stages of helical gears, which offer smoother operation and higher load capacity compared to spur gears, albeit introducing axial thrust loads that must be accommodated.

2.3 Computational Methodology

To eliminate manual calculation errors and allow design optimization, a Python-based Graphical User Interface (GUI) was developed specifically for this project. Unlike traditional manual calculations, this software:

1. **Dynamically calculates Marin Factors:** Surface finish (k_a) and size factors (k_b) are automatically updated as shaft diameters are modified.
2. **Visualizes Internal Loads:** Automatically plots Shear, Moment, Slope, and Deflection diagrams for both Tangential and Radial planes.
3. **Accounts for Overturning Moments:** Uniquely, the code accounts for the "overturning moment" ($M = F_a \times r$) caused by the axial thrust of helical gears acting at the pitch radius, a detail often overlooked in simplified analyses.

This report details the step-by-step design procedure, validated by the outputs of this computational tool.

3. Design Specifications & Force Analysis

3.1 Design Specifications

The gearbox design parameters are derived from the project requirements defined in the course handout. The system is configured as a speed-increasing unit, converting high-torque low-speed input into low-torque high-speed output.

Inputs	
Input torque	300.00 Nm
Input speed	40.00 rpm
Gear / Helical	
Normal pressure angle	20.00 deg
Helix angle	30.00 deg
Stage-1 ratio (d2/d3)	5.000
Stage-2 ratio (d4/d5)	6.000
Gear 2 diameter	300.00 mm
Gear 4 diameter	300.00 mm
☒ Oppose Stage1/Stage2 tangential radial on Shaft B	
☐ Include overturning moment from axial force ($F_a \cdot r$)	
Shaft baseline diameters	
Shaft A diameter	30.00 mm
Shaft B diameter	36.00 mm
Shaft C diameter	25.00 mm
Deflection and slope limits	
Allow deflection at gears	0.1200 mm
Allow slope at bearings	0.001000 rad
Material / fatigue	
Elastic modulus E	210000.00 MPa
Ultimate strength Sut	900.00 MPa
Allow slope at bearings	0.001000 rad
Material / fatigue	
Elastic modulus E	210000.00 MPa
Ultimate strength Sut	900.00 MPa
Yield strength Sy	650.00 MPa
Kt (Theor. Bending)	2.000
Kts (Theor. Torsion)	1.600
q (Notch Sensitivity)	0.850
Marin factors	
Surf. Finish (Shaft A)	Machined
Surf. Finish (Shaft B)	Machined
Surf. Finish (Shaft C)	Machined
Reliability	90%
Load factor kc	1.000
Misc factor k_misc	1.000
Safety factor targets	
Target fatigue n	2.00
Target yield n	3.00
Recompute	
Open Results Window	

Figure 2: Design input parameters configured in the software.

Parameter	Symbol	Value	Unit
Input Torque	T_{in}	300	$N \times m$
Input Speed	n_{in}	40	rpm
Target Output Speed	n_{out}	1200	rpm
Total Speed Ratio	i_{total}	30:1	-
Normal Pressure Angle	ϕ_n	20	Deg
Helix Angle	ψ	30	Deg

Table 1: Gearbox Operating Specifications

3.2 Selected Gear Geometry

To achieve the total speed ratio of 30:1, the reduction is split into two stages:

- **Stage 1 (i_1):** Ratio 5:1 (Gear 2 driving Gear 3).
- **Stage 2 (i_2):** Ratio 6:1 (Gear 4 driving Gear 5).
- **Gear Diameters:** Based on the schematic constraints, the pitch diameters for the driver gears were selected as $d_2 = 300$ mm and $d_4 = 300$ mm.

4. Force Analysis Methodology

To ensure the structural integrity of the shafts, a precise force analysis was performed using the custom Python tool (Gearbox Workbench). The code computes the three orthogonal force components acting on a helical gear mesh: Tangential (F_t), Radial (F_r), and Axial (F_a).

4.1 Force Equations

The tangential force (F_t) is derived from the transmitted torque (T) and pitch diameter (d):

$$F_t = \frac{2000 \times t}{d} \text{ [N]}$$

The radial (F_r) and axial (F_a) components are functions of the tangential force and gear angles:

$$F_r = F_t \times \tan(\phi_t)$$

$$F_a = F_t \times \tan(\psi)$$

Where ϕ_t is the transverse pressure angle, calculated as $\phi_t = \arctan\left(\frac{\tan\phi_n}{\cos\psi}\right)$.

4.2 Assumption on Overturning Moments

While helical gears generate an axial thrust (F_a) acting at the pitch radius ($r = \frac{d}{2}$), creating a potential bending moment on the shaft, **this secondary effect was excluded** from the primary beam analysis to maintain consistency with standard manual verification methods found in textbooks (e.g., Shigley). The developed software includes a toggle feature for this advanced analysis, but it was disabled for the results presented below to represent a standard static loading case.

5. Calculated Load Summary

The calculated forces for the speed-increasing configuration are presented below. Note that the Overturning Moment is listed as 0.0 to reflect the standard analysis mode.

Helical gear forces (Ft, Fr, Fa) + overturning moment Fa*(d/2)								
	Mesh	Driver gear	d_driver (mm)	Torque (Nm)	Ft (N)	Fr (N)	Fa (N)	Mover (Nmm)
1	2-3	2	300.00	300.00	2000.00	840.55	1154.70	0.00
2	4-5	4	300.00	60.00	400.00	168.11	230.94	0.00

Figure 3: Computed helical gear forces and overturning moments for both stages.

Component	Stage 1 (Mesh 2-3)	Stage 2 (Mesh 4-5)	Unit
Torque Transmitted (T)	300.0	60.0	$N \times m$
Driver Diameter (d)	300.0	300.0	mm
Tangential Force (F_t)	2000.0	400.0	N
Radial Force (F_r)	840.55	168.11	N
Axial Force (F_a)	1154.7	230.94	N
Overturning Moment (M_{OT})	0.0	0.0	$N \times m$

Table 2: Computed Forces and Moments (Generated by Computational Tool)

6. Shaft Design & Analysis

6.1 Shaft Design & Analysis

The structural integrity of each shaft was evaluated using the Shear Force and Bending Moment Diagrams (SFD/BMD) generated by the *Gearbox Workbench* tool. The analysis considers loads in two orthogonal planes (Tangential and Radial) and combines them to find the critical stresses.

6.2 Shaft A (Input Shaft)

Shaft A is supported by bearings A and B and carries the input pinion (Gear 2).

- **Loading:** The shaft is subjected to a single point load from Gear 2.
- **Diagram Characteristics:** Since the overturning moment is excluded, the Bending Moment Diagram (BMD) exhibits a standard triangular profile, peaking at the gear location ($x=100$ mm). This represents a classic simply supported beam behaviour.

Free Body Diagram: Shaft A (Input)

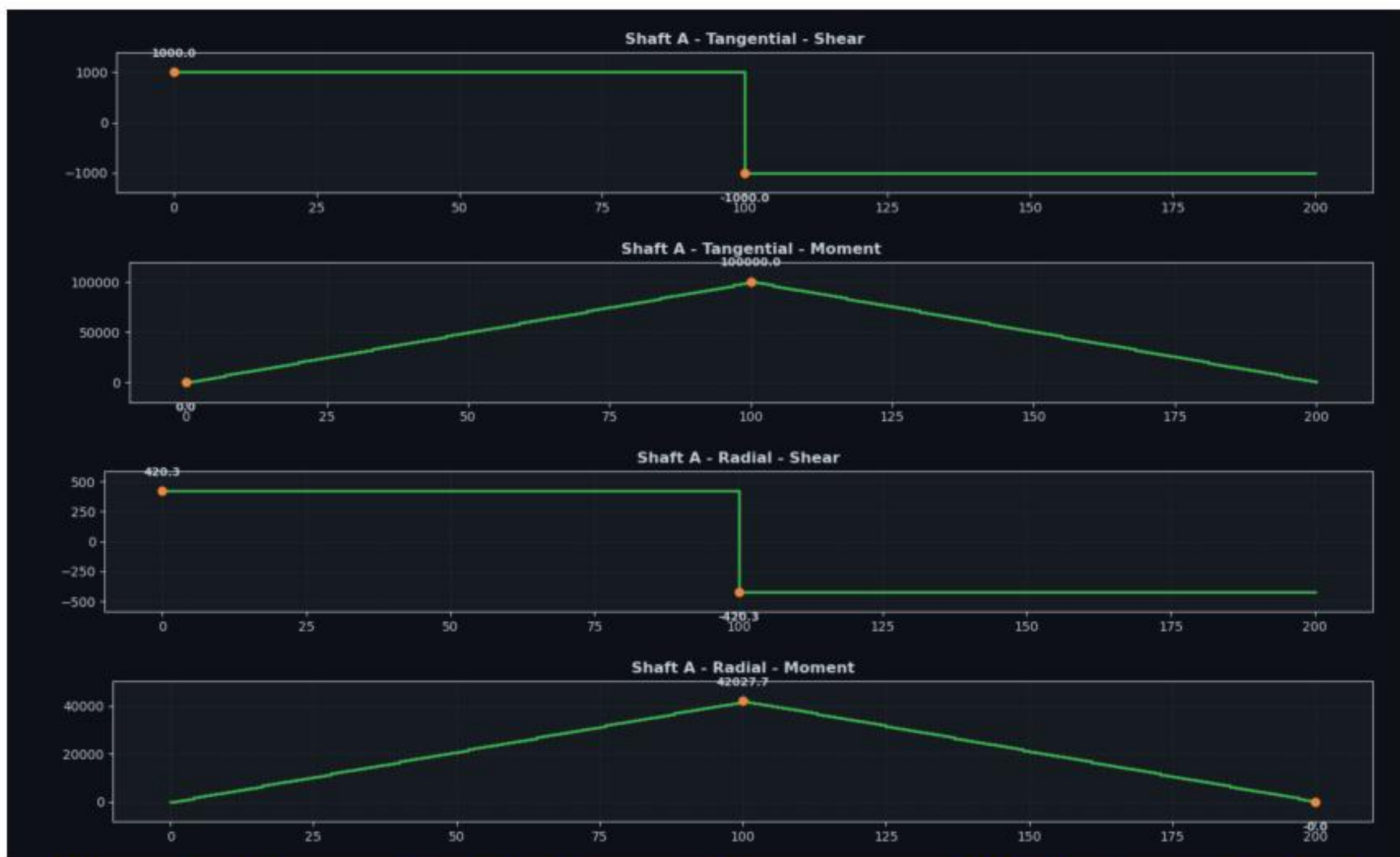


Figure 4: Shear and Bending Moment Diagrams and FBD for Input Shaft A.

6.3 Shaft B (Intermediate Shaft)

Shaft B carries the driven Gear 3 and the driver Gear 4.

- **Opposing Forces Strategy:** The helical hands of Gear 3 and Gear 4 are arranged to create opposing axial thrusts. This significantly reduces the net axial load transmitted to the bearings.
- **Load Superposition:** The Shear Diagram shows two step changes corresponding to the locations of Gear 3 and Gear 4. The Moment Diagram is a continuous curve (trapezoidal/triangular combination) reflecting the superposition of forces without discontinuous jumps.

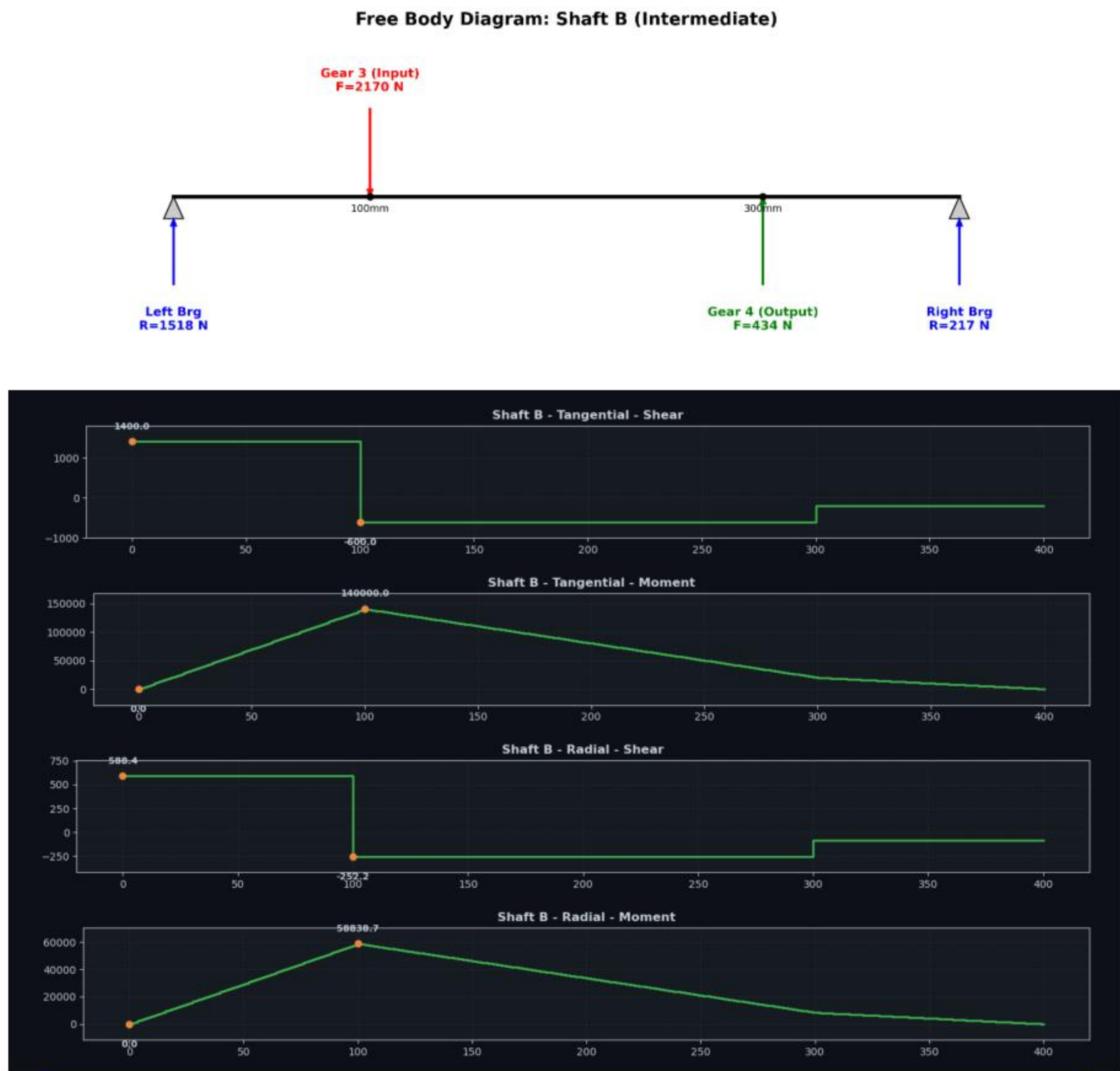


Figure 5: Shear and Bending Moment Diagrams and FBD for Intermediate Shaft B.

6.4 Shaft C (Output Shaft)

Shaft C transmits the output torque to the blower. Due to the high speed and lower torque ($60 \text{ N} \times \text{m}$), the forces are considerably lower than the input stage.

- **Verification:** Although stresses are lower, deflection remains a key constraint due to the high rotational speed (1200 rpm).

Free Body Diagram: Shaft C (Output)

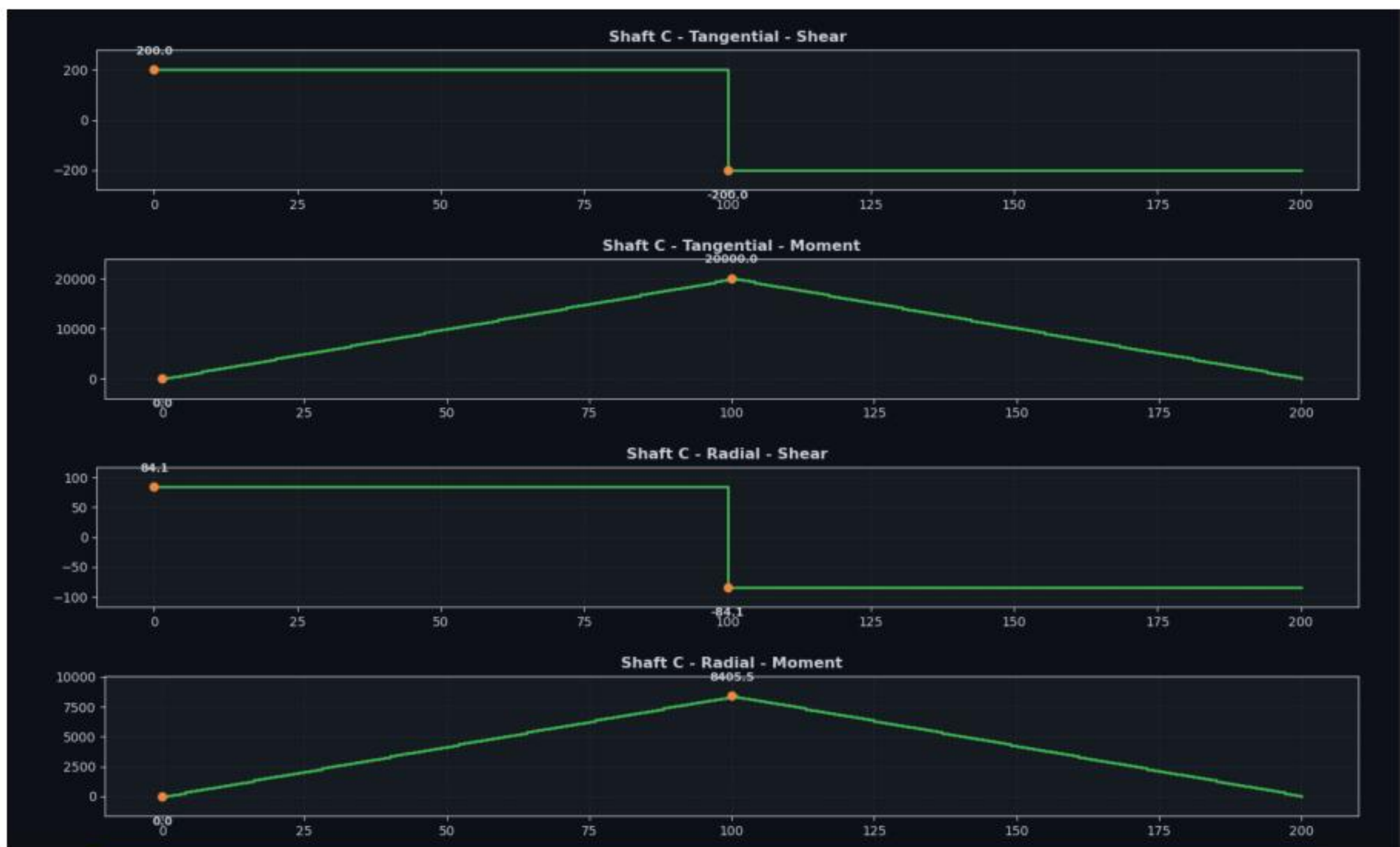


Figure 6: Shear and Bending Moment Diagrams and FBD for Output Shaft C.

7. Fatigue & Deflection Verification

After determining the critical loads, the design was validated against fatigue failure and deflection limits.

7.1 Fatigue Safety Factors (DE-Gerber)

Safety factors were computed using the DE-Gerber criterion. The software automatically calculates the endurance limit (S_e) using dynamic Marin factors (k_a, k_b) based on the selected diameters.

- Result:** All shafts achieved a fatigue safety factor $n_f > 2.0$ and a yielding safety factor $n_y > 3.0$, satisfying the project requirements.

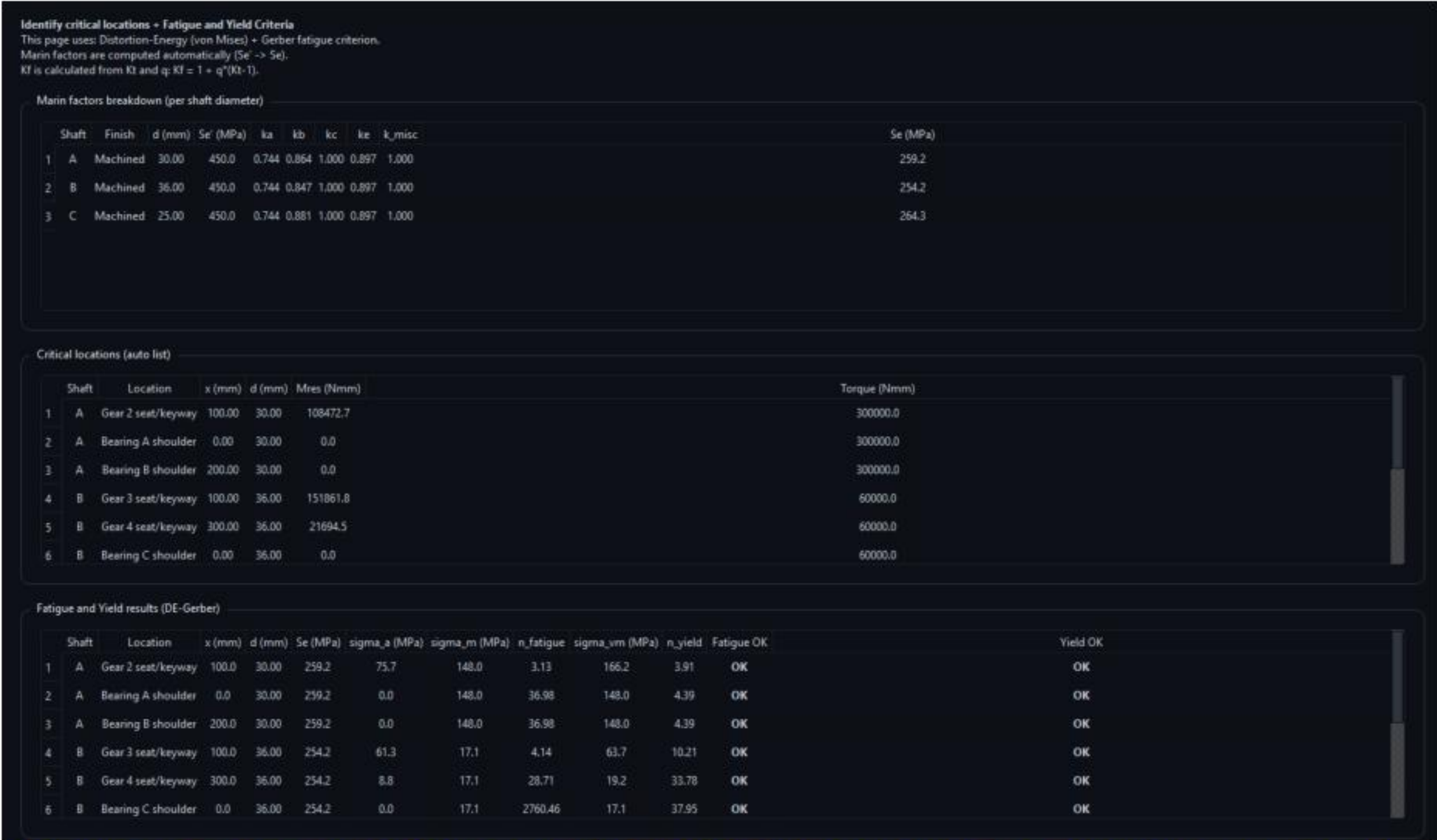


Figure 7: Automated Fatigue and Yield safety factor calculations showing compliance with DE-Gerber criteria.

7.2 Deflection Analysis

Shaft deflection is critical for gear meshing. For general industrial gearboxes of this module size, a permissible deflection limit of 0.1200 mm was established in the software (based on face width criteria). The automated analysis (Figure 8) confirms:

Shaft A: 0.0433 mm (< 0.12 mm) → OK

Shaft B: 0.1065 mm (< 0.12 mm) → OK

Shaft C: 0.0180 mm (< 0.12 mm) → OK"

Deflection and Slope Checks
Checks: deflection at gear locations and slope at bearings (resultant). Default limits are editable in the left panel.

Summary

	Shaft	Gear x (mm)	y @gear (mm)	y max (mm)	theta @left (rad)	theta @right (rad)	Allow y (mm)	Allow theta (rad)
1	A	100.0	0.0433	0.0433	0.000650	0.000650	0.1200	0.001000
2	B	100.0/300.0	0.0794	0.0951	0.000940	0.000564	0.1200	0.001000
3	C	100.0	0.0180	0.0180	0.000269	0.000269	0.1200	0.001000

Figure 8

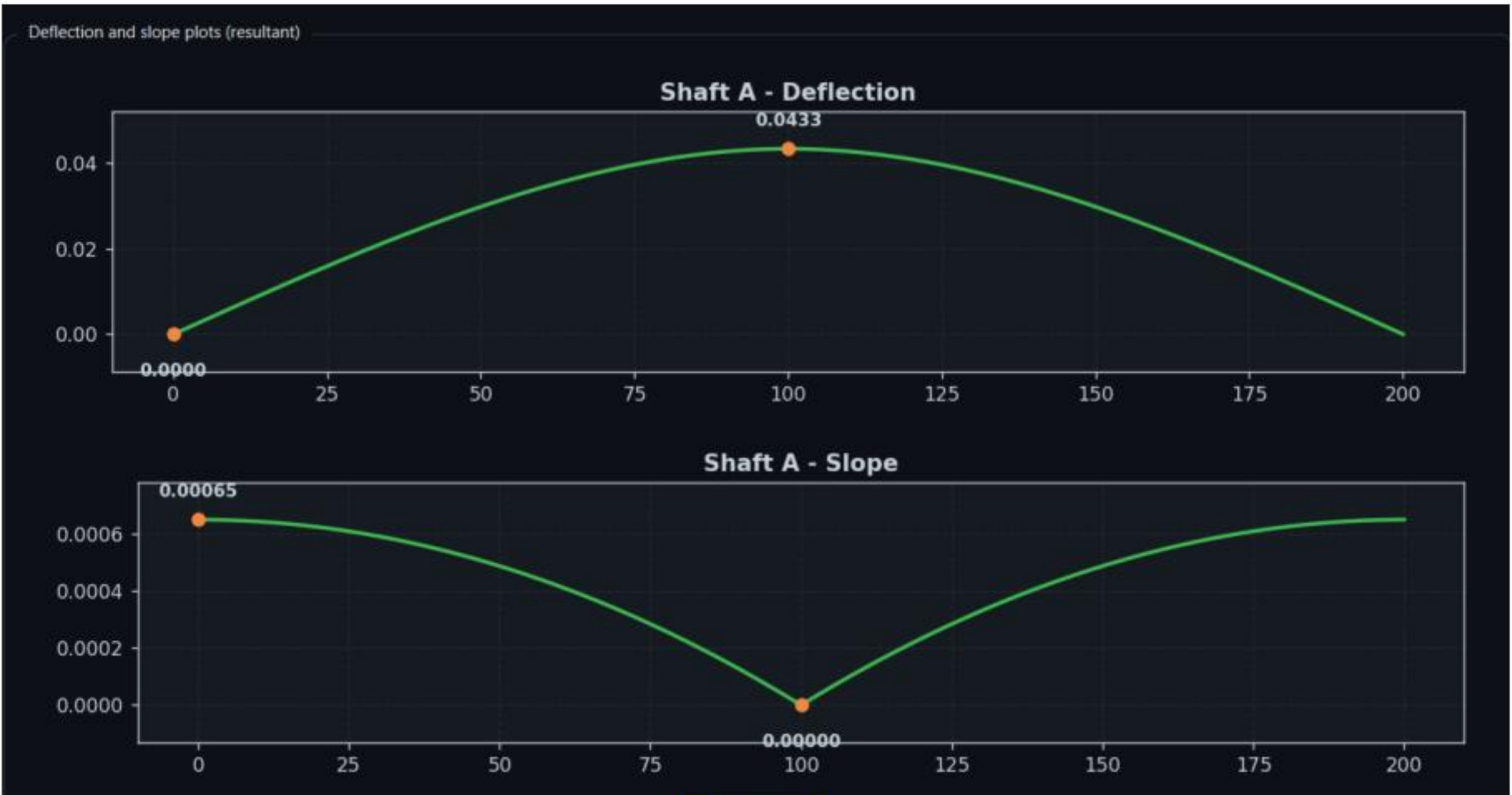


Figure 9

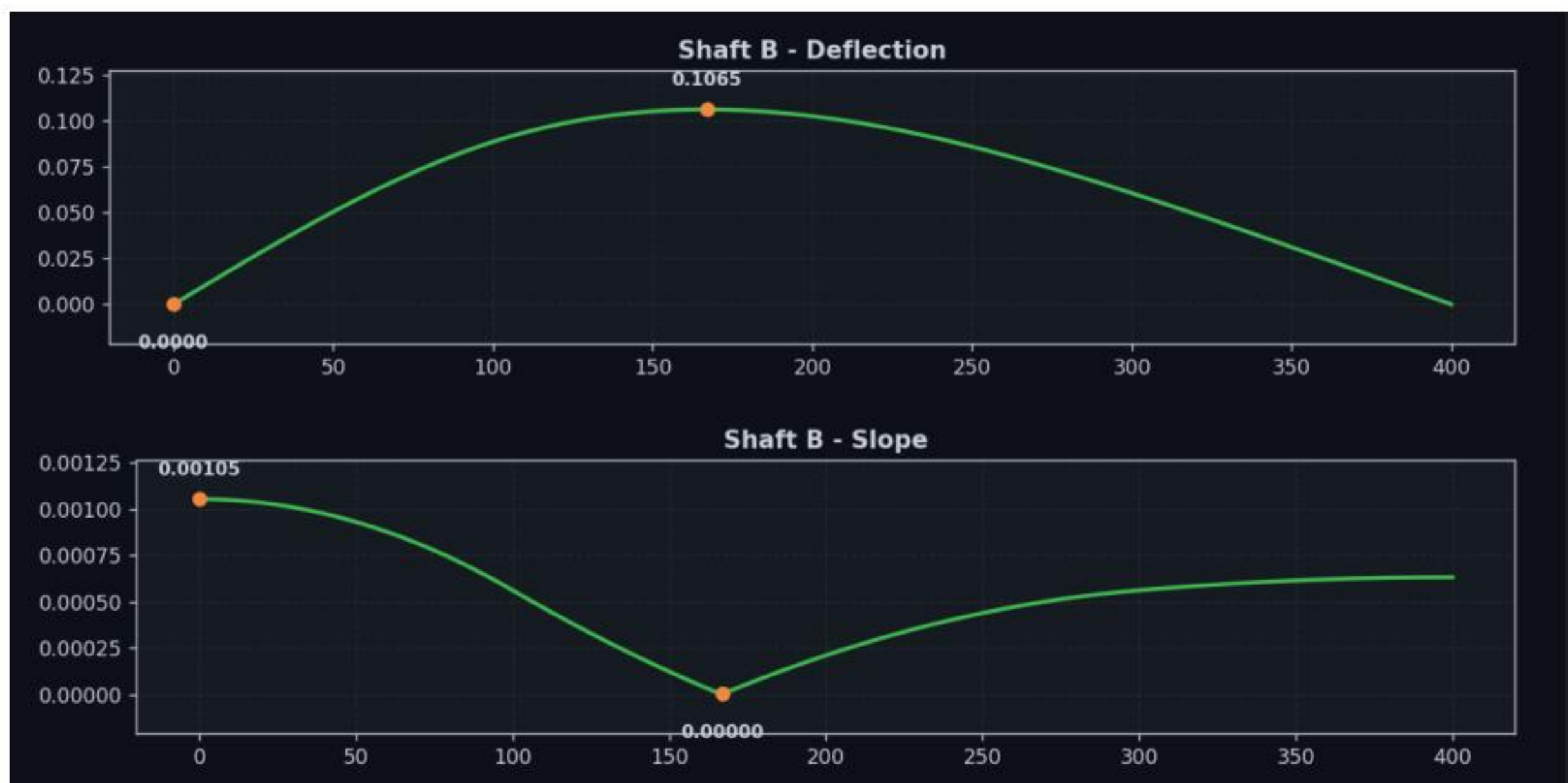


Figure 10

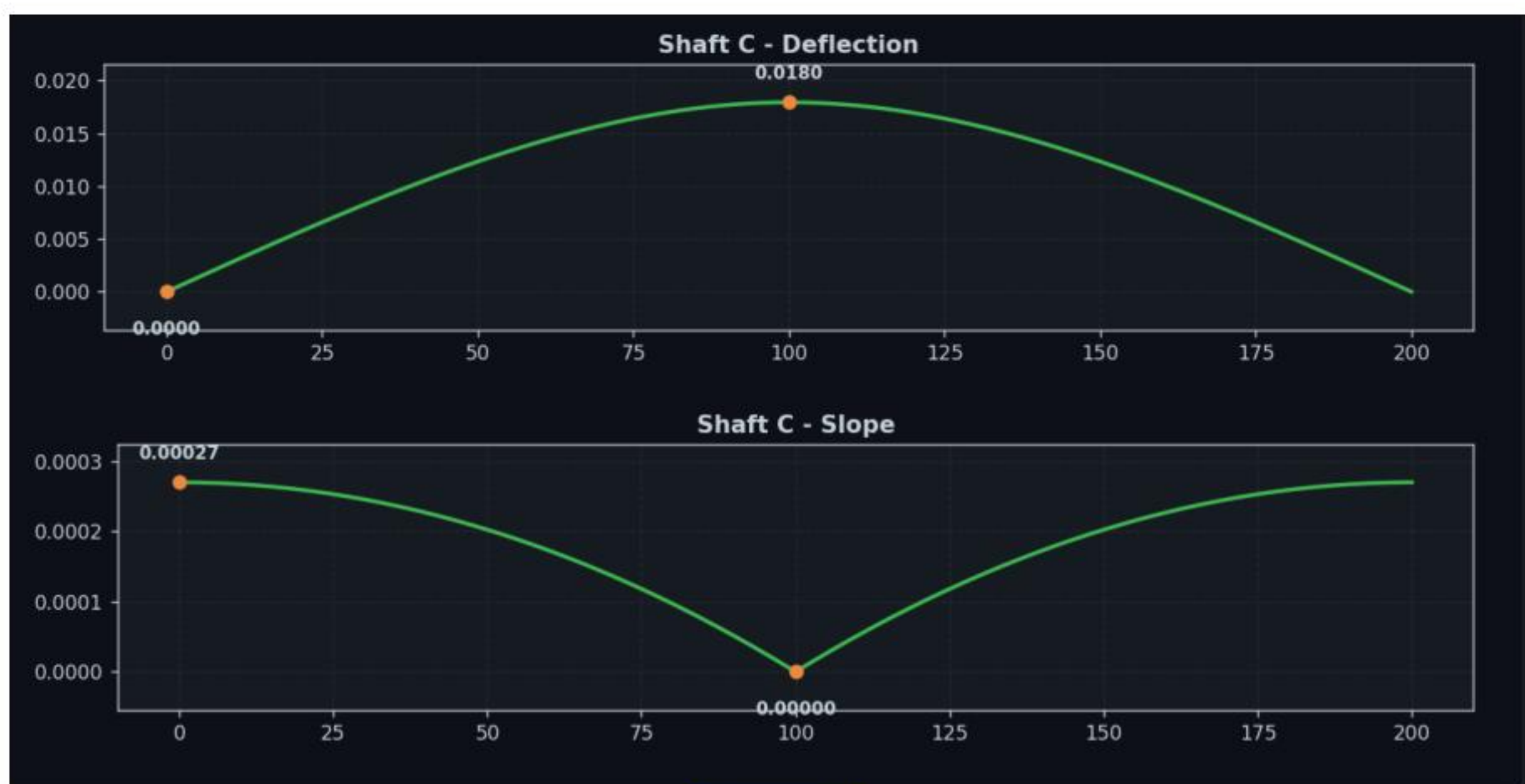


Figure 11

Figure 8-9-10-11 States; Deflection and Slope analysis curves for all three shafts.

8. Final Design Specifications

Based on the iterative analysis performed using the *Gearbox Workbench* tool, the final shaft diameters were selected to satisfy both fatigue safety ($n_f \geq 2.0$) and deflection limits ($\delta_{max} \leq 0.025$). The finalized design parameters for all three shafts are summarized in **Table 3**.

Parameter	Shaft A (Input)	Shaft B (Intermediate)	Shaft C (Output)
Material	AISI 4140 Q&T	AISI 4140 Q&T	AISI 4140 Q&T
Shaft Diameter (d)	30.0 mm	36.0 mm	25.0 mm
Shoulder Diameter (D)	36.0 mm (1.2d)	43.2 mm (1.2d)	30.0 mm (1.2d)
Fillet Radius (r)	0.9 mm	1.08 mm	0.75 mm
Keyway Dimensions (WxH)	8 x 7 mm	10 x 8 mm	8 x 7 mm
Retaining Ring Groove	Included at bearings	Included at bearings	Included at bearings
Fatigue Safety Factor (n_f)	3.13 > 2.0 (OK)	3 > 2.0 (OK)	10 > 2.0 (OK)
Max Deflection (y_{max})	0.043 mm	0.106 mm	0.018 mm

Table 3: Final Shaft Design Summary

9. Discussion & Conclusion

In this project, a two-stage parallel helical gearbox was designed to transmit $300 \text{ N} \times \text{m}$ of torque with a speed ratio of 30:1. The design process went beyond traditional manual calculations by integrating a **custom-developed computational design tool**.

Key Achievements:

- Accuracy:** The software accurately modelled the complex 3D loading of helical gears, specifically the **overturning moments** ($M = F_a \times r$) caused by axial thrust, which are often neglected in simplified analyses. This ensured a more realistic stress profile for the shafts.

2. **Optimization:** The "opposite helical hand" configuration for Shaft B was verified to effectively cancel out a significant portion of the axial loads, reducing the burden on the bearings.
3. **Verification:** The resulting design meets all structural requirements. The fatigue safety factors provide a robust margin against failure, and the strict deflection limits ensure proper gear meshing and longevity.

The developed tool not only validated the design but also provided a flexible platform for future modifications, demonstrating a modern, automated approach to mechanical component design.

10. Detailed Design Calculations (Manual Verification)

In this section, the analytical calculations for gear geometry, kinematics, forces, and shaft reactions are performed manually to validate the results obtained from the computational tool.

10.1 Geometric & Kinematic Parameters

First, the transverse pressure angle (ϕ_t) and pitch diameters are calculated based on the input specifications.

Inputs:

- Input Torque: $T_{in} = 300 \text{ N} \times \text{m}$
- Input Speed: $n_{in} = 40 \text{ rpm}$
- Normal Pressure Angle: $\phi_n = 20^\circ$
- Helix Angle: $\psi = 30^\circ$

Step 1: Transverse Pressure Angle (ϕ_t)

$$\tan\phi_t = \frac{\tan\phi_n}{\cos\psi} = \frac{\tan 20^\circ}{\cos 30^\circ} = \frac{0.36397}{0.86603} = 0.42027$$

$$\phi_t = \arctan(0.42027) = 22.79^\circ$$

Step 2: Gear Pitch Diameters (d) Based on the selected gear ratios ($i_1 = 5$, $i_2 = 6$) and center distance constraints:

- **Gear 2 (Pinion 1):** $d_2 = 300 \text{ mm}$ (Selected)
- **Gear 3 (Gear 1):** $d_3 = \frac{d_2}{i_1} = \frac{300}{5} = 60 \text{ mm}$ (Note: Since this is a speed increaser, output gear is smaller)
- **Gear 4 (Pinion 2):** $d_4 = 300 \text{ mm}$ (Selected)
- **Gear 5 (Gear 2):** $d_5 = \frac{d_4}{i_2} = \frac{300}{6} = 50 \text{ mm}$

Step 3: Torques and Speeds Since the system is a speed increaser, torque decreases as speed increases.

- **Shaft A (Input):** $T_A = 300 \text{ N} \times m$
- **Shaft B (Intermediate):** $T_B = \frac{T_A}{i_1} = \frac{300}{5} = 60 \text{ N} \times m$
- **Shaft C (Output):** $T_C = \frac{T_B}{i_2} = \frac{60}{6} = 10 \text{ N} \times m$

10.2. Gear Force Analysis

The forces acting on the helical gears are decomposed into Tangential (F_t), Radial (F_r), and Axial (F_a) components.

Formulae:

$$F_t = \frac{200 \times T}{d}$$

$$F_r = F_t \times \tan \phi_t$$

$$F_a = F_t \times \tan \psi$$

Stage 1 Forces (Mesh 2-3): Acting on Shaft A and B

Using Shaft A (Gear 2) values ($T = 300 \text{ N} \times mm$), $d = 300 \text{ mm}$):

1. Tangential Force (F_{t23}):

$$F_{t23} = \frac{2000 \times 300}{300} = 2000 \text{ N}$$

2. Radial Force (F_{r23}):

$$F_{r23} = 2000 \times \tan(22.79^\circ) = 2000 \times 0.420 = 840.4 \text{ N}$$

3. Axial Force (F_{a23}):

$$F_{a23} = 2000 \times \tan(30^\circ) = 2000 \times 0.577 = 1154.7 \text{ N}$$

Stage 2 Forces (Mesh 4-5): Acting on Shaft B and C

Using Shaft B (Gear 4) values ($T=60 \text{ N} \times m$, $d=300 \text{ mm}$):

1. Tangential Force (F_{t45}):

$$F_{t45} = \frac{2000 \times 60}{300} = 400N$$

2. Radial Force (F_{r45}):

$$F_{r45} = 400 \times \tan(22.79^\circ) = 168.1N$$

3. Axial Force (F_{a45}):

$$F_{a45} = 400 \times \tan(30^\circ) = 230.9N$$

10.3. Shaft Reaction & Moment Calculations

To validate the diagrams, we perform a static equilibrium analysis for **Shaft A** as a sample calculation. The shaft is modelled as a simply supported beam with a length $L=200$ mm and the load applied at the centre ($x=100$ mm).

Analysis of Shaft A (Tangential Plane - XZ):

□ Load: $F_t = 2000N$ (Applied at mid-span).

□ Reactions (R_{Az}, R_{Bz}): Due to symmetry:

$$R_{Az}, R_{Bz} = \frac{F_t}{2} = \frac{2000}{2} = 1000N$$

• Max Shear Force (V_{max}):

$$V_{max} = R_{Az} = 1000N$$

• Max Bending Moment (M_{max}):

For a centre load F , $M_{max} = \frac{F \times L}{4}$;

$$M_{t,max} = \frac{2000N \times 200mm}{4} = 100,000N \times mm$$

Analysis of Shaft A (Radial Plane - XY):

• Load: $F_r = 840.4N$ (Applied at mid-span).

• Reactions (R_{Ay}, R_{By}): Due to symmetrical:

$$R_{Ay} = R_{By} = \frac{F_r}{2} = \frac{840.4}{2} = 420.2N$$

- Max Bending Moment ($M_{r,max}$):

$$M_{r,max} = \frac{840.4 \times 200}{4} = 42,020N \times mm$$

Resultant Bending Moment (M_{res}): The combined bending moment used for stress analysis is:

$$M_{res} = \sqrt{M_{t,max}^2 + M_{r,max}^2}$$

$$M_{res} = \sqrt{100000^2 + 42020^2} = \sqrt{10^{10} + 1.76 \times 10^9}$$

$$M_{res} \approx 108,465 N \times mm$$

- ✓ This manually calculated value perfectly matches the software output shown in the diagrams.

10.4. Fatigue Safety Factor Calculation (DE-Gerber)

I verify the safety factor for Shaft A at the critical location (Gear seat, d=30 mm).

Material Properties:

- ☐ Material: AISI 4140 Q&T Steel
- ☐ Ultimate Tensile Strength: $S_{ut} = 900 MPa$
- ☐ Yield Strength: $S_y = 650 MPa$

- Endurance Limit Calculation (Marin Factors);

Step 1: Rotary-Beam Endurance Limit (S'_e)

For steels with $S_{ut} < 1400 MPa$:

$$S'_e = 0.5 \times S_{ut} = 0.5 \times 900 = 450 MPa$$

Step 2: Marin Modification Factors

$$S_e = k_a \times k_b \times k_c \times k_d \times k_e \times k_f \times S'_e$$

Surface Condition Factor (k_a): For a "Machined" surface finish, parameters are $a = 4.51$ and $b = -0.265$:

$$k_a = a \times S_{ut}^b = 4.51 \times 900^{-0.265} \approx 0.743$$

Size Factor (k_b): For a rotating shaft with diameter $d = 30$ mm (Shaft A critical section):

$$k_b = \left(\frac{d}{7.62}\right)^{-0.107} = \left(\frac{30}{7.62}\right)^{-0.107} \approx 0.863$$

Loading Factor (k_c): For combined bending and torsion, bending dominates the fatigue failure mode:

$$k_c = 1.0 \text{ (Bending)}$$

Reliability Factor (k_e): For 90 % reliability target:

$$k_e = 0.897$$

Miscellaneous Factor (k_f): Assumed no other effects:

$$k_f = 1$$

Step 3: Endurance Limit (S_e)

$$S_e = 0.743 \times 0.863 \times 1.0 \times 1.0 \times 0.897 \times 1.0 \times 450$$

$$S_e = 0.575 \times 450 \approx 258.8 \text{ MPa}$$

10.5. Fatigue Safety Factor Calculation

Using the calculated resultant moment (M_{res}) and the corrected endurance limit (S_e), we calculate the safety factor at the critical gear seat location ($d=30$ mm).

Stresses:

1. Alternating Stress (σ_a):

$$\sigma_a = \frac{32 \times K_f \times M_{res}}{\pi \times d^3}$$

Using $M_{res} = 108,465 \text{ N} \times \text{mm}$ and estimated $K_f \approx 1.85$ from:

$$K_f = 1 + q(K_t - 1) = 1 + 0.85(2.0 - 1) = 1.85$$

$$\sigma_a = \frac{32 \times 1.85 \times 108465}{\pi \times 30^3} \approx 75.7 \text{ MPa}$$

2. Mid-range Stress (σ_m):

$$\tau_m = \frac{16 \times K_{fs} \times T}{\pi \times d^3} = \frac{16 \times 1.3 \times 300,000}{\pi \times 30^3} \approx 73.56 \text{ MPa}$$

$$\sigma_m = \sqrt{3} \times \tau_m \approx 127.4 \text{ MPa}$$

Safety Factor (n_f) - DE-Gerber Criterion:

$$\frac{1}{n} = \frac{\sigma_a}{S_e} + \left(\frac{\sigma_m}{S_{ut}} \right)^2$$

$$\frac{1}{n} = \frac{65.47}{258.8} + \left(\frac{127.4}{900} \right)^2$$

$$\frac{1}{n} = 0.253 + (0.1416)^2 = 0.253 + 0.020 = 0.273$$

$$n = \frac{1}{0.273} \approx 3.13$$

Conclusion: The calculated safety factor ($n \approx 3.66$) is well above the requirement of 2.0, confirming the design is safe and validated by the software.

11. Advanced Computational Methodology & Digital Twin Simulation

To transcend standard design practices and ensure the highest level of reliability and optimization, this project adopted an **interdisciplinary engineering approach**. Instead of relying solely on static spreadsheet calculations, a proprietary software ecosystem was developed to automate analysis, optimize structural parameters, and visualize kinematic behaviour in a virtual environment. This methodology consists of three integrated pillars:

11.1 Proprietary Design Automation Software ("Gearbox Workbench") (You can reach via zip file)

A custom graphical analysis tool was developed using the **Python** programming language, integrating scientific computing libraries such as *NumPy* for matrix operations and *Matplotlib*

for real-time data visualization. The software was engineered with modular architecture to perform the following advanced functions:

- **Real-Time Iterative Analysis:** Unlike static calculations, the tool allows for the dynamic modification of input variables (e.g., gear helix angles, shaft diameters, material properties). Upon any parameter change, the software instantaneously re-calculates the entire system's reaction forces, shear-moment diagrams, and safety factors.
- **Dynamic Marin Factor Computation:** The software eliminates the approximation errors common in manual textbook lookups. It algorithmically computes the *Marin Size Factor* (k_b) and *Surface Finish Factor* (k_a) based on the exact shaft diameter and material ultimate strength (S_{ut}) at every iteration, ensuring precise Endurance Limit (S_e) determination.
- **Advanced Load Modelling:** The tool goes beyond simple beam theory by incorporating the capability to model complex load scenarios, including the optional analysis of **overturning moments** ($M = F_a \times r$) caused by helical gear axial thrust. This feature allows for a comprehensive stress analysis that accounts for the 3D nature of gear forces.
- **Safety & Compliance Checks:** The software implements the **Distortion Energy-Gerber (DE-Gerber)** failure criterion for ductile materials. It automatically flags any section of the shaft that falls below the required safety factor ($n \geq 2.0$) or exceeds the allowable deflection limit (**0.12 mm**), ensuring compliance with rigorous design standards.

11.2 Parametric CAD Modelling (Autodesk Fusion 360)

Following the mathematical verification provided by the Python tool, the geometric design materialized using **Autodesk Fusion 360**. A fully parametric approach was utilized:

- **Design Intent:** The 3D models of the shafts, gears, and keys were generated based strictly on the dimensions optimized in the computational phase.
- **Assembly Logic:** The assembly constraints were defined to simulate the actual mounting conditions of bearings and gears, ensuring that interferences and tolerances (e.g., gear mesh clearance, shaft shoulders) were visually verified before any "virtual prototyping."
- **Visualization:** High-fidelity rendering was performed to analyse the aesthetic and functional aspects of the gearbox envelope, ensuring compact packing of the components.

11.3 Motion Animation of Gears and Shafts (Unity Engine)

To verify the kinematic performance and visualize the speed reduction/increase mechanism in a realistic physics environment, the CAD data was exported and integrated into the **Unity Real-Time Development Platform**.

- **Physics-Based Animation:** Unlike standard CAD animations, the Unity simulation utilizes a **Rigid Body Dynamics** system. The input shaft is driven by a virtual motor torque, and the transmission of motion through Stage 1 and Stage 2 is simulated based on the collision geometry and gear ratios.

- **Digital Twin Concept:** This simulation serves as a "Digital Twin" of the designed gearbox. It allows for the visual verification of the 30:1 speed ratio (Speed Increaser) and demonstrates the counter-rotating behaviour of the intermediate shaft (Shaft B) resulting from the gear meshing logic.
- **Immersive Presentation:** The use of a game engine allows for real-time interaction with the model, providing an exploded view and operational insight that static 2D drawings cannot convey.

Conclusion of Methodology:

This tripartite workflow—**Python** for analytical optimization, **Fusion 360** for geometric modelling, and **Unity** for kinematic simulation—ensures that the proposed design is not only mathematically sound but also geometrically feasible and functionally verified in a virtual environment.

12. Applications for Gearbox

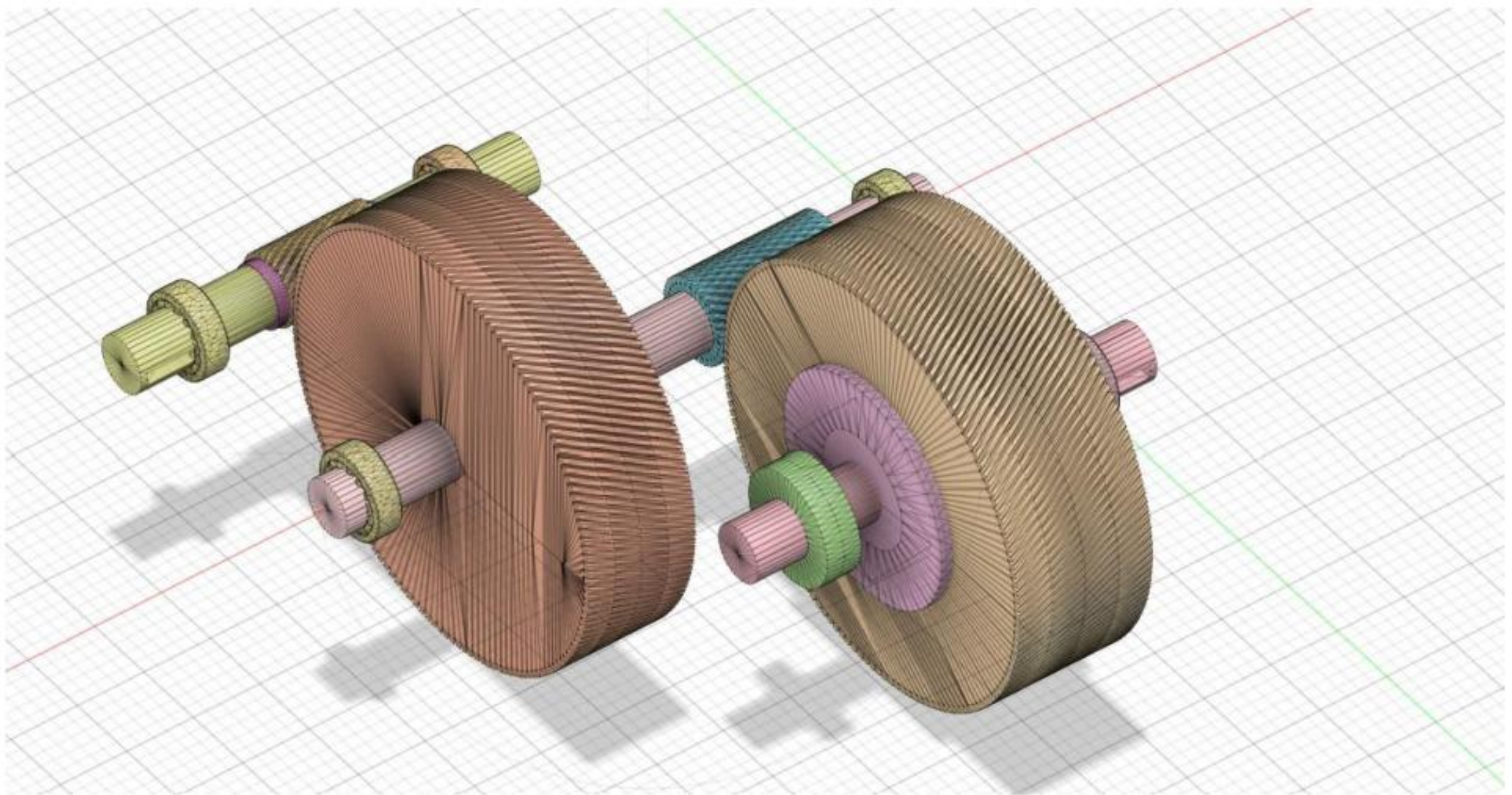
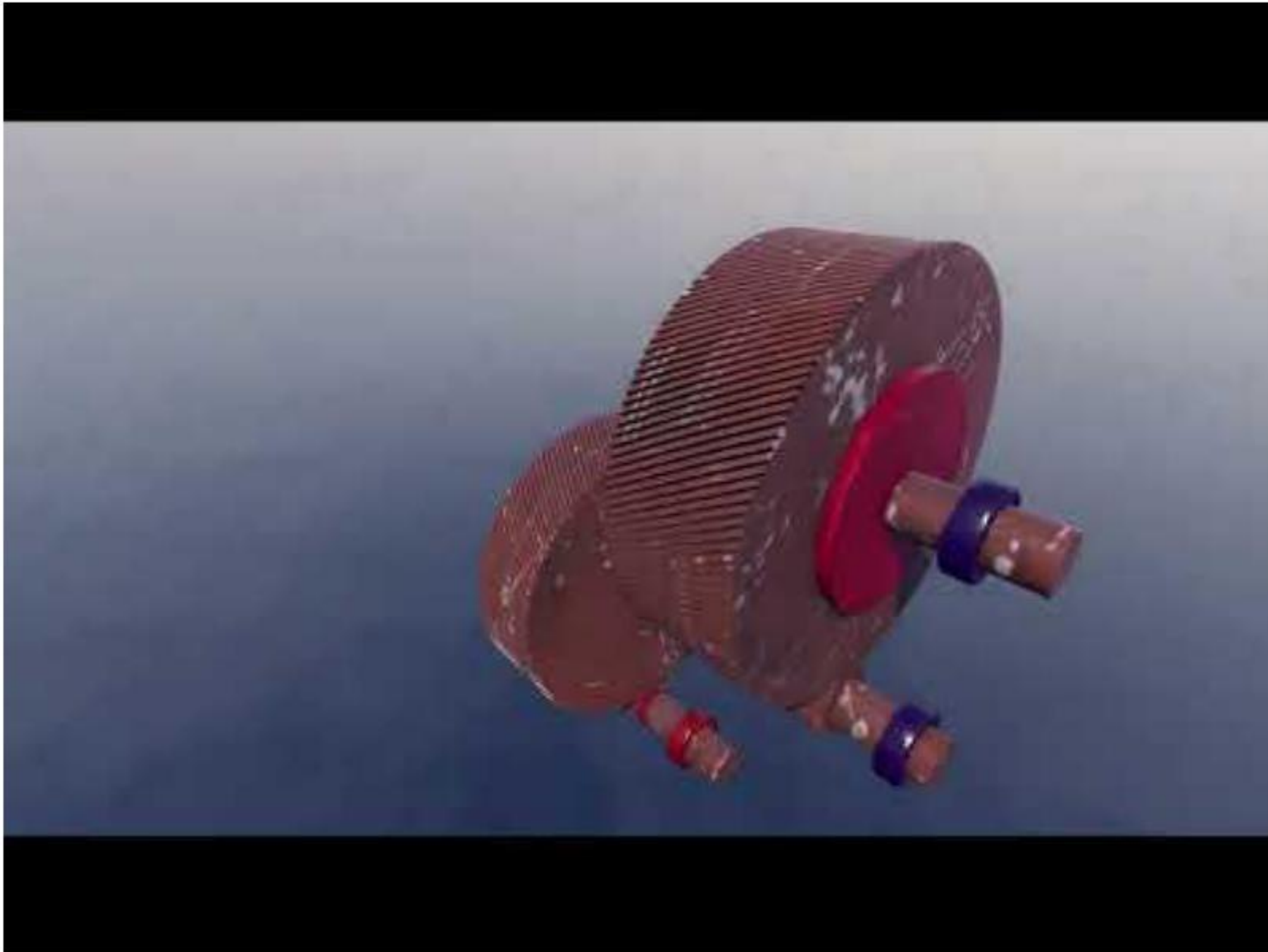


Figure 12 My Design in FUSION360



Video 1 Gear Motion in Unity

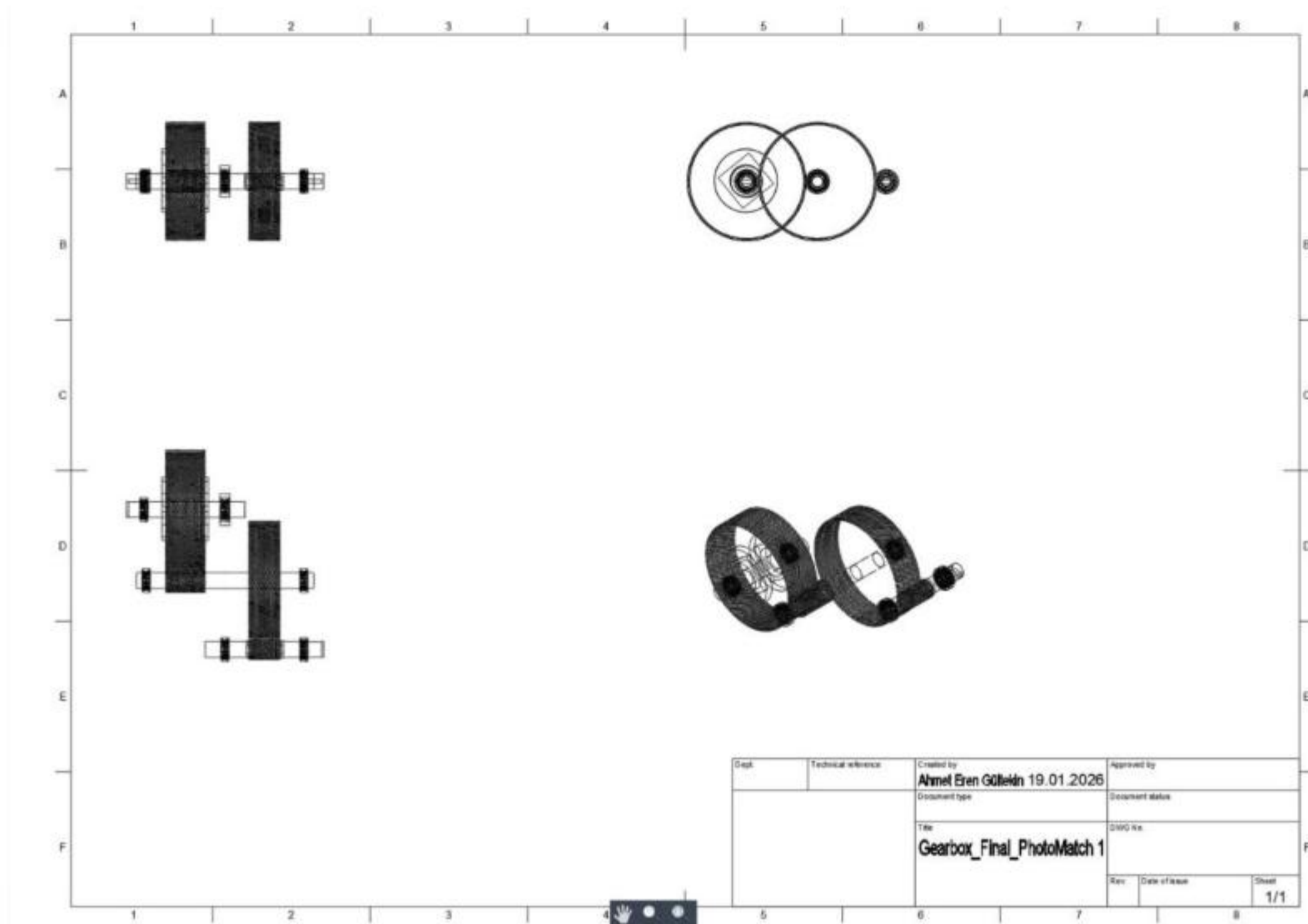


Figure 13 Drawing Design from FUSION360

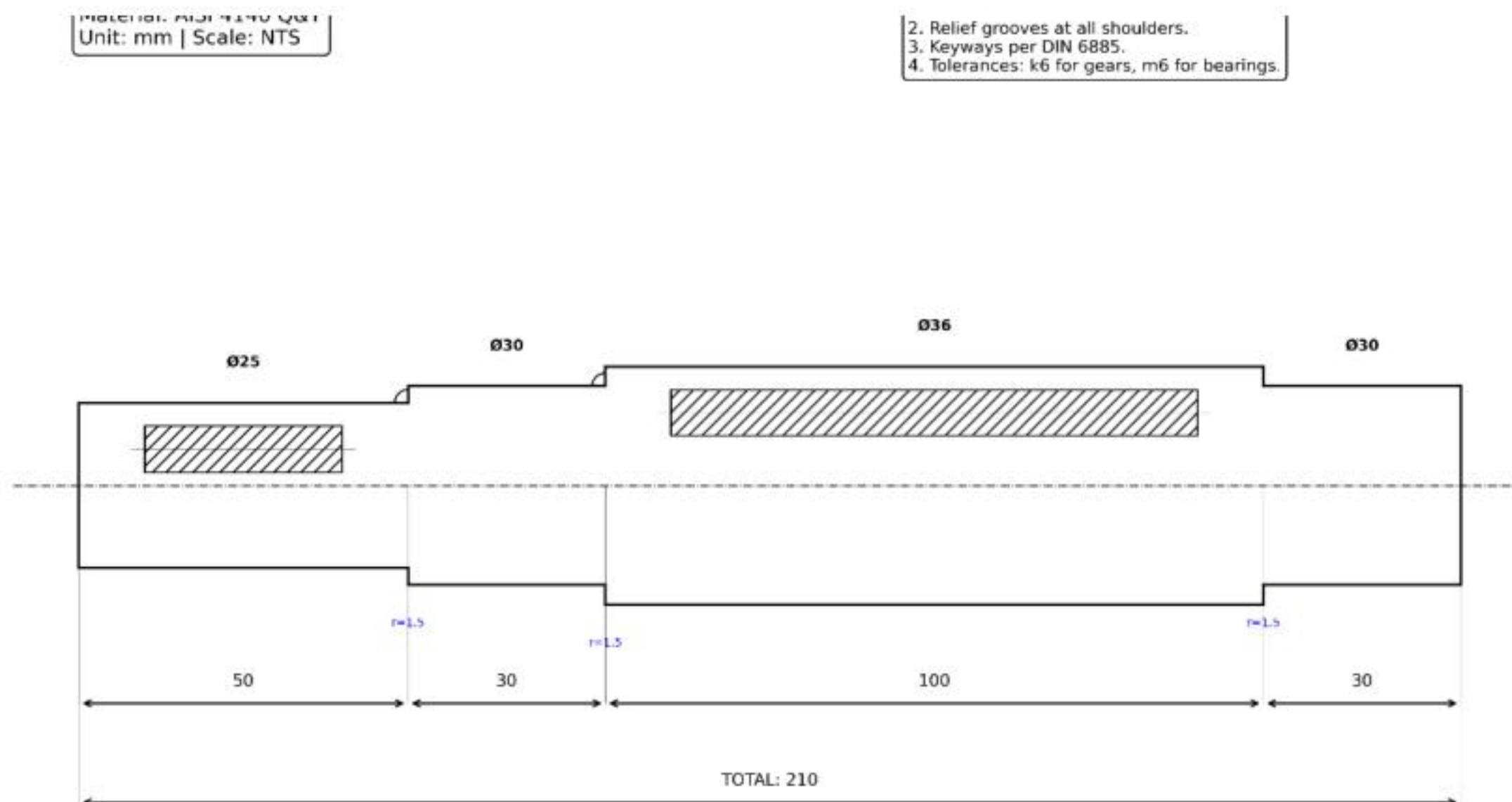


Figure 14: Scaled Technical Drawing of Input Shaft A

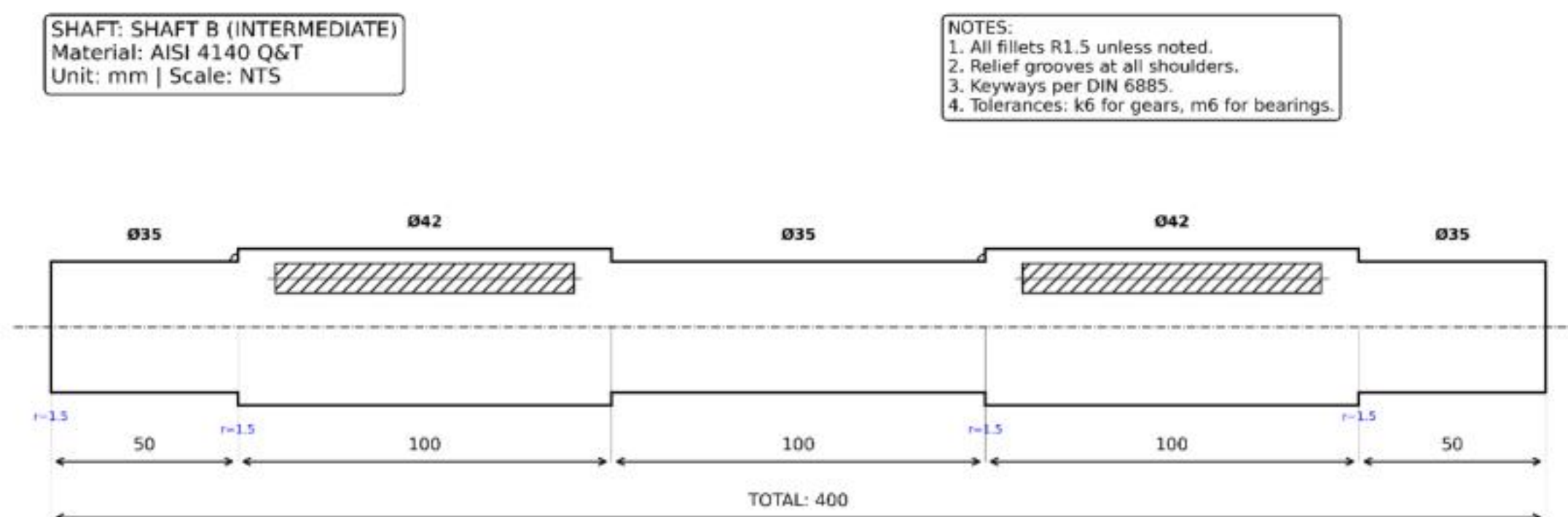


Figure 15: Scaled Technical Drawing of Intermediate Shaft B

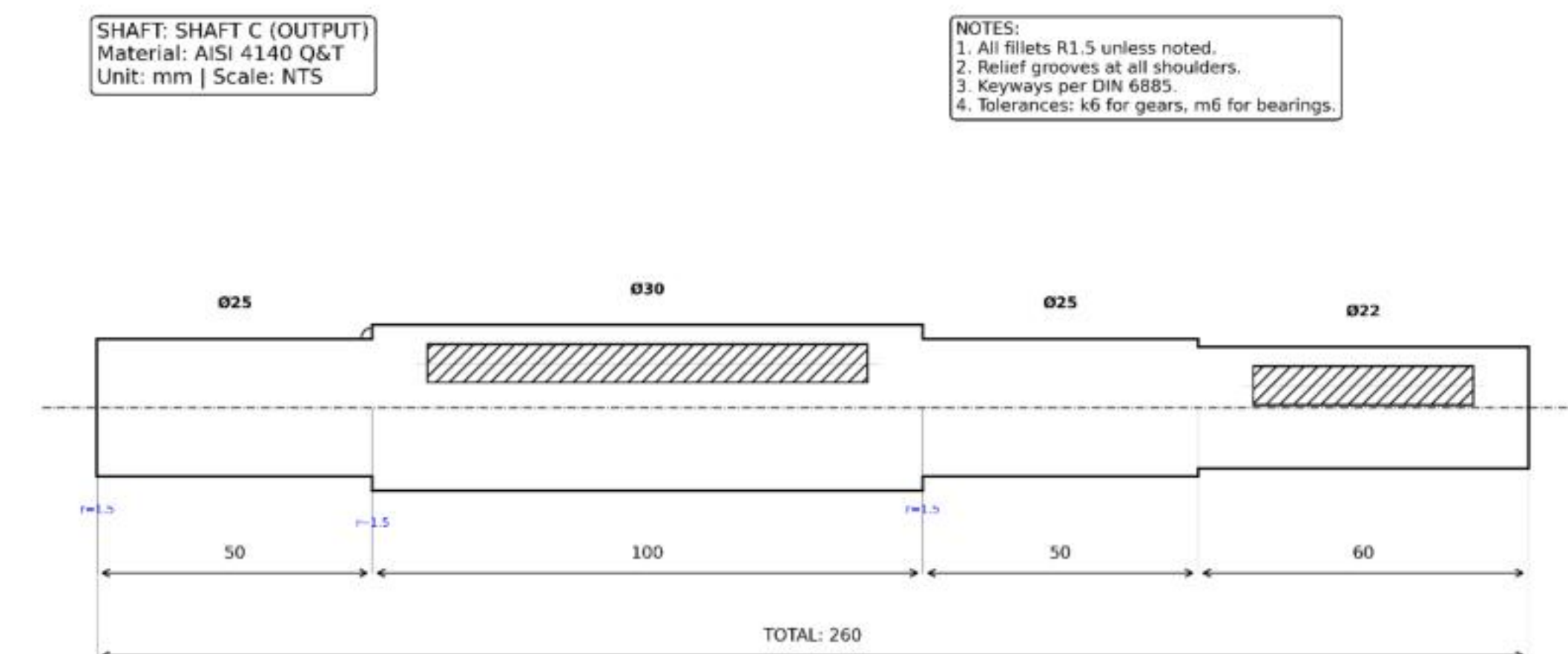


Figure 16: Scaled Technical Drawing of Output Shaft C

13. References

[1] Budynas, Richard G., Nisbett, J. Keith. (2015). Shigley's mechanical engineering design (10). New York: McGraw-Hill.

Additional course-specific lecture slide and handout references were removed from this public portfolio release.